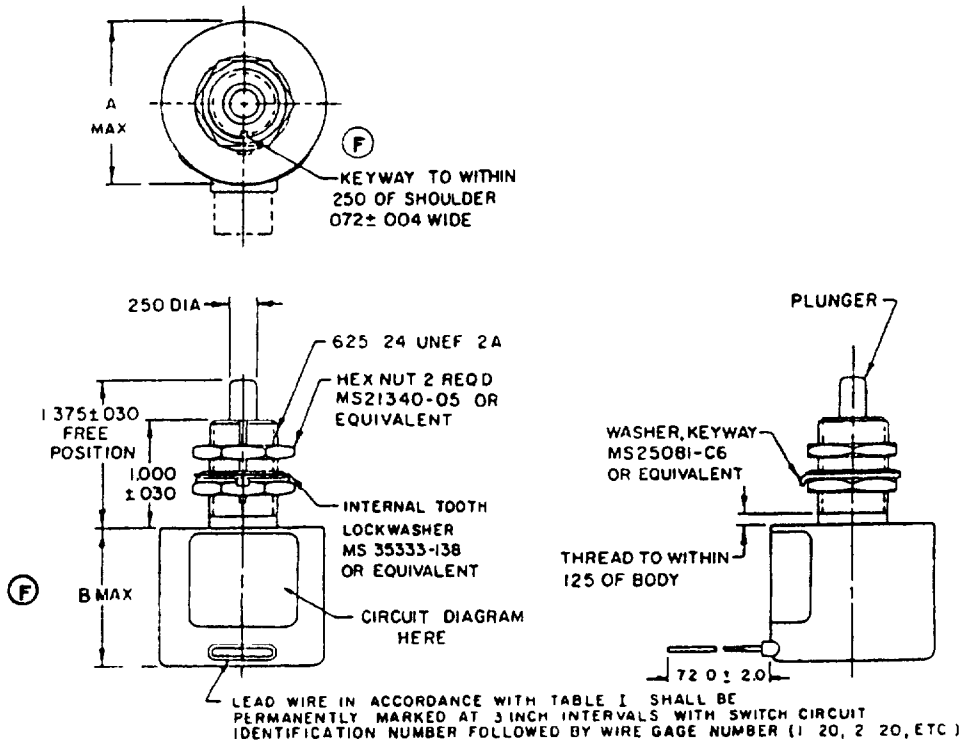
MILITARY SPECIFICATION SHEET

SWITCHES, SENSITIVE, PLUNGER, RESILIENT SEAL, FLUID RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-8805.



INCHES	MM	INCHES	MM
004	0 10	250	6 35
015	0 38	584	14 83
.017	0.43	.625	15 88
022	0.56	650	16 51
030	0.76	.870	22 10
038	0 97	1.000	25 40
072	1.83	1.375	34 93
.125	3.18	2.0	50.80
		72 0	1828 00

NOTES
1 Dimensions are in inches
2 Unless otherwise specified, tolerance is ± 010 (0 25 mm)
3 Metric equivalents are given for general information only
4 Contour optional, provided dimensions specified are not exceeded

FIGURE 1 Configuration and dimensions

(F) denotes changes

FSC 5930

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TABLE I. Part numbers and characteristics.

Part number	A dia max	B max	Lead wires 1/	
			Size	No. req'd
MS24331-1	1.015 (25.78)	.980 (24.89)	No. 20	6
MS24331-2 2/	1.515 (38.48)	1.500 (38.10)	No. 18	6
MS24331-3	1.515 (38.48)	1.310 (33.27)	No. 18	4
MS24331-4	1.515 (38.48)	1.700 (43.18)	No. 18	8
MS24331-6	1.015 (25.78)	1.000 (25.40)	No. 18	6

1/ Lead wire shall be MIL-W-22759/7.
2/ Inactive for new design.

REQUIREMENTS

Dimensions and configuration See figure 1 and table I.

Enclosure design. 4 (Resilient). 1/

Temperature characteristic 1 (-55°C to +85°C).

Shock type M (100G).

Vibration grade 1 (10 to 500 Hz).

Maximum weight with leads.

MS24331-1 .46 pound.

MS24331-2 .77 pound.

(F) MS24331-3 .62 pound.

(F) MS24331-4 .84 pound.

(F) MS24331-6 .60 pound.

Operating characteristics

Actuating force: 9 ±3 pounds.

(F) Full overtravel force 30 pounds maximum.

Release force

MS24331-1 4 pounds minimum.

MS24331-2, -3, -4, and -6: 5 pounds minimum.

Pretravel: .070 inch maximum.

Movement differential.

MS24331-1: .020 inch maximum.

MS24331-2, -3, and -6: .035 inch maximum.

(F) MS24331-4 .045 inch maximum.

Overtravel .250 inch minimum

1/ All entrances to the switch cavity except through the actuator bushing shall be sealed by fusion of glass-to-metal, metal-to-metal, or ceramic-to-metal and the lead wires shall be potted to provide stress relief.

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Coincidence of operating and releasing points: All circuits shall transfer within .010 inch of plunger travel after first circuit transfers.

Strength of actuating means. 30 pounds.

Material.

Potting. Shall meet the hydrolytic stability requirements of MIL-I-16923 or MIL-S-23586, as applicable.

Plastic -2 General purpose mineral and flock filled phenolics may be used internally.

Finish: Switch housing shall be processed to resist corrosion.

Contact resistance Not applicable.

Dielectric withstanding voltage·

At atmospheric pressure· 1,000 V rms.

At reduced barometric pressure. 50,000 feet, 400 V rms.

Following electrical endurance· Points of application between all unconnected terminals of the sample pole - not applicable.

Mechanical endurance. 25,000 cycles.

Electrical endurance· 25,000 cycles.

Electrical ratings See table II.

Fluid resistance Except for the cut end of the lead wire, switches shall be submerged in each of the following fluids for 2 to 2-1/2 minutes, which shall consist of one cycle (one cycle is 10 to 12-1/2 minutes total). Each switch shall be subjected to three cycles.

a. Turbine fuel (MIL-T-5624).

b. Hydraulic fluid (MIL-H-83306).

c. Coolanel (FSN 9150-551-4022).

d. Ethylene glycol (MIL-E-9500).

e. Lubricating oil (MIL-L-7808).

After each immersion, the excess fluid shall be blown off the external surfaces of the switch with an air jet. Following the third cycle, the switch shall be subjected to and shall meet the requirements for dielectric withstanding voltage, insulation resistance, operating characteristics, and seal tests.

Part number. See table I.

TABLE II. Electrical ratings.

Load	Sea level 28 V dc (amperes)				50,000 feet 28 V dc (amperes)			
	-1	-2, -6	-3	-4	-1	-2, -6	-3	-4
Resistive	4	10	15	15	4	10	15	15
Inductive	2	3	10	10	2	3	10	10
Motor	4	6	5	5	4 1/	6 1/	5 1/	5 1/

1/ Application information only.

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Qualification inspection.

Group submission See table III.

Group A inspection:

Seal test Only watertight test shall be performed.

TABLE III. Qualification inspection (group submission).

Examination or test	Samples	Extent of approval
Qualification inspection table of MIL-S-8805 1/	MS24331-1 (26 units)	All
Visual and mechanical examination Groups II and V	MS24331-2, -3, -4, -6 (2 units each)	
Group VII	MS24331-3, -6 (10 units each)	

1/ Two sample units shall be tested for fluid resistance following visual and mechanical examination of group I.

NOTE Part number MS24331-5, covered by MIL-S-8805/40A, is canceled. For future procurement use part number MS24331-1.

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities.
Army - AR
Navy -AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - MI, AV
Navy -MC
Air Force - 19

Preparing activity.
Navy - EC

Agent
DLA - ES

(Project 5930-1314)

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